

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

Course Name: Cloud Computing and Big Data Analytics

Course Code: CSA1505

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I P.Abdul – 192525426(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Abdul

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

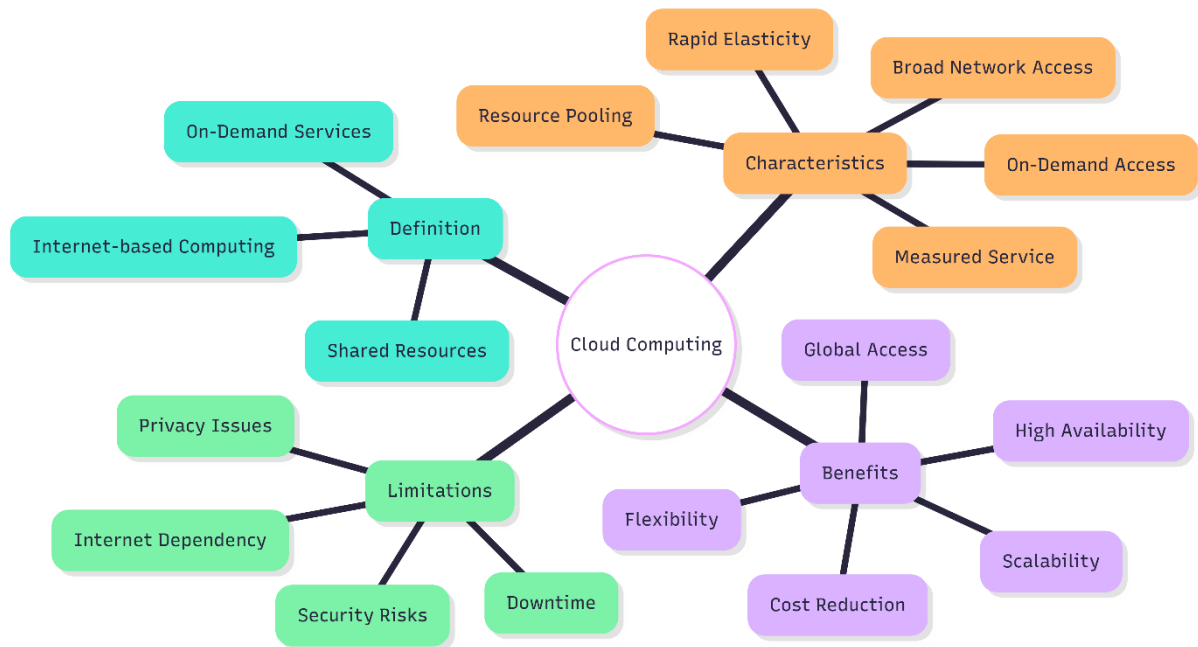
- Elasticity, scalability, resource pooling, on-demand access, measured service

- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

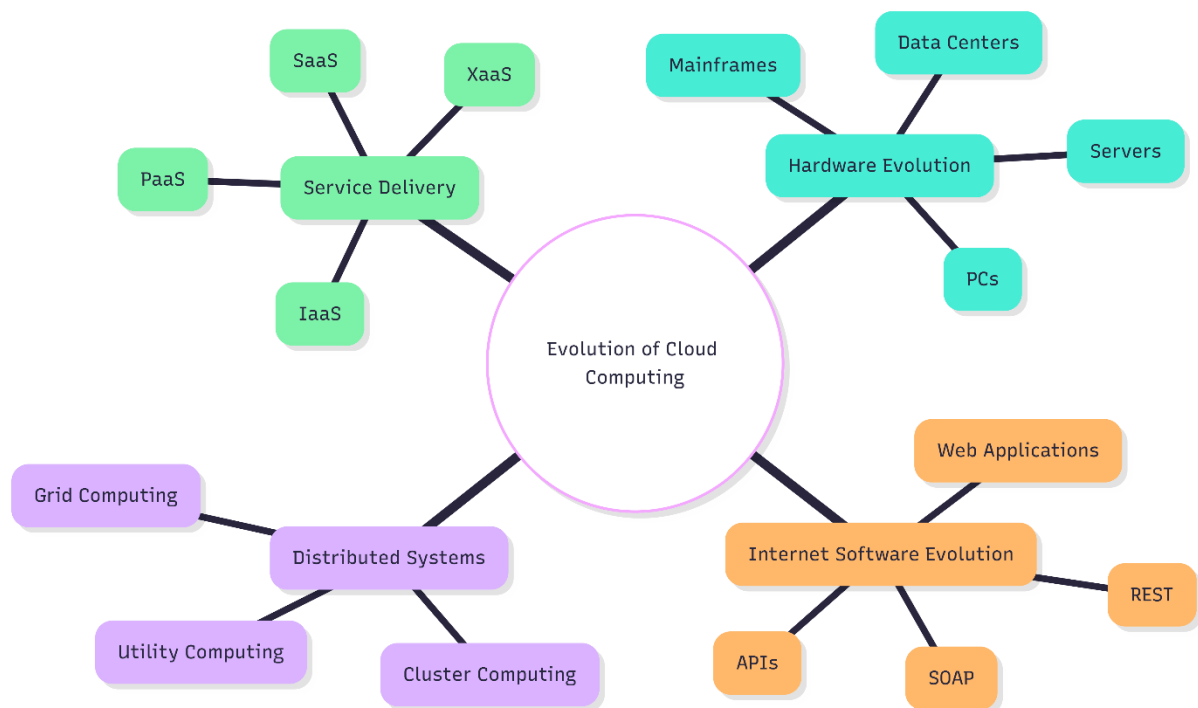
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

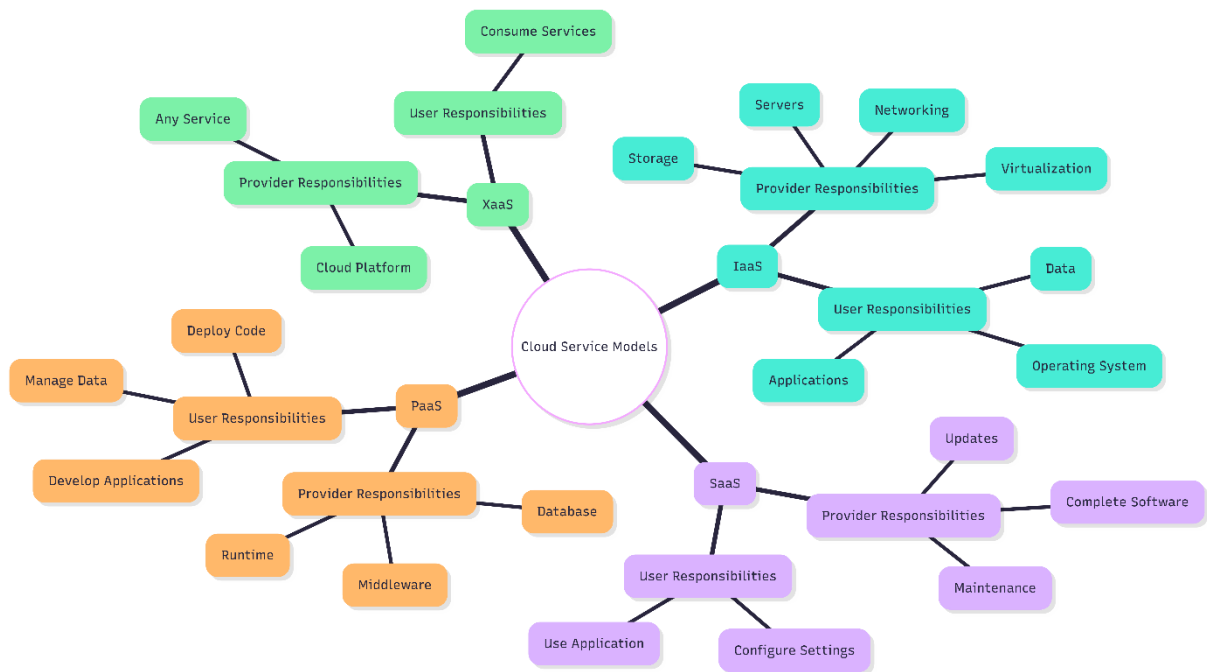
Tool Used: Mermaid
Concept Map 1:



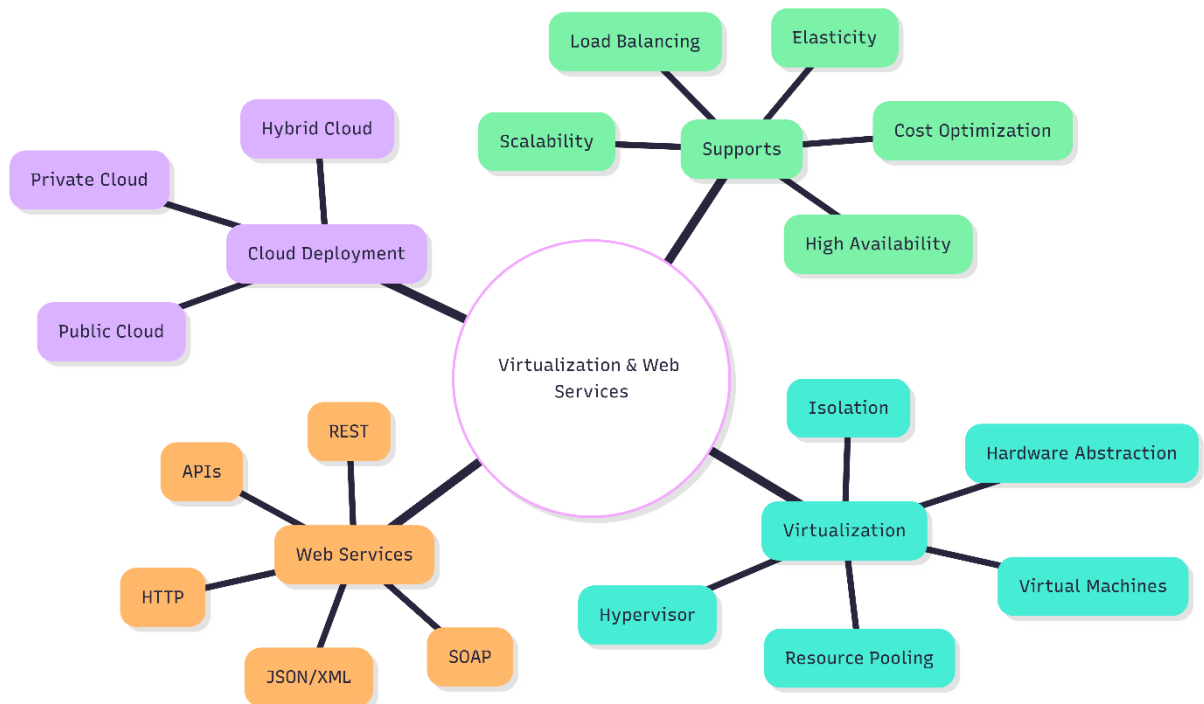
Concept Map 2:



Concept Map 3:



Concept Map 4:



Task 4: Technical Justification

The concept maps effectively demonstrate the fundamental principles of cloud computing by organizing key concepts into logical hierarchies and showing meaningful relationships among cloud characteristics, service models, virtualization, web services, and cloud evolution. The diagrams illustrate how cloud technologies work together to provide scalable, flexible, and efficient computing services. The inclusion of service models (IaaS, PaaS, SaaS, and XaaS), APIs, virtualization, elasticity, scalability, and shared responsibility clearly satisfies the Course Outcome (CO1) by explaining the foundations of cloud computing architecture and service delivery.

1. Evaluate the effectiveness of your concept map in representing cloud computing principles.

The concept map effectively represents cloud computing principles by organizing concepts from general to specific. The central node identifies the main topic, while branches clearly illustrate cloud characteristics, service models, virtualization, and web services. Relationship labels such as **supports**, **provides**, **enables**, and **depends on** improve readability and explain how different cloud components interact. Colour coding and hierarchical organization make the diagram easier to understand and accurately represent cloud computing concepts.

2. Analyse how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability allows cloud resources to expand as user demand increases, while elasticity automatically allocates or releases resources based on workload changes. Resource pooling enables multiple users to share computing resources efficiently, reducing operational costs and improving performance. Without these capabilities, applications could experience slow performance, resource shortages, increased infrastructure costs, service interruptions, and poor user experience during peak workloads.

3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

The concept map includes **APIs**, **SOAP**, and **REST** because they enable communication between cloud services and client applications. REST APIs provide lightweight, scalable communication suitable for modern web applications, while SOAP offers secure and standardized communication for enterprise systems. The selected cloud service models (IaaS, PaaS, SaaS, and XaaS) demonstrate different levels of managed services, allowing users to choose solutions based on their infrastructure, development, or software requirements. These technologies improve interoperability, automation, flexibility, and overall system efficiency.

4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The relationships between virtualization, web services, cloud service models, and cloud characteristics illustrate how cloud systems operate efficiently. Virtualization enables multiple virtual machines to share physical hardware, improving resource utilization and reducing hardware costs. APIs and web services allow seamless communication between providers and consumers. Elasticity and scalability ensure consistent performance under varying workloads, while shared responsibility clearly defines the security and management roles of cloud providers and users. Together, these interactions improve system reliability, application performance, and user experience.

5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Creating the concept map improved my understanding of how cloud computing components are interconnected rather than functioning independently. It helped me recognize the relationships between cloud characteristics, service models, virtualization, APIs, and cloud deployment. If redesigning the map, I would include deployment models (Public, Private, Hybrid, and Community Cloud), additional real-world examples such as AWS, Microsoft Azure, and Google Cloud Platform, and more cross-links between concepts to better illustrate their interactions and dependencies.